

Riley Greiff, P.E.

AI / Machine Learning Engineer

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SUMMARY

Machine learning engineer who ships LLM-powered products end to end — from data pipeline to deployed, user-facing app. MS in Computational Analytics from Georgia Tech and a licensed Professional Engineer (TN) with 7 years as a transportation engineer solving quantitative, high-stakes problems. Strengths in applied LLM systems (retrieval, extraction, ranking), production ML serving, and inference efficiency.

TECHNICAL SKILLS

Languages: Python, SQL, R, TypeScript

ML / AI: PyTorch, Transformers, LightGBM, FAISS, RAG, two-tower retrieval, learning-to-rank, quantization

LLM / Applied: Claude API (Haiku/Sonnet), structured extraction, agent loops, cost-aware routing, offline eval

Systems / MLOps: FastAPI, Docker, Redis, PostgreSQL, Prometheus, dynamic batching, load testing, CI

Data / Web / Cloud: pandas, NumPy, A/B testing & bootstrap CIs, Next.js/React, Vercel, Google Cloud, Neon, Stripe

EDUCATION

M.S., Computational Analytics — Georgia Institute of Technology

2023 – 2025

B.S., Civil Engineering — University of Tennessee, Knoxville

2015 – 2019 · GPA 3.79/4.00

SELECTED PROJECTS

The Wedding Dossier — Live AI Product *Next.js, PostgreSQL, Claude API, Google Cloud | theweddingdossier.com*

- Built and operate a live production LLM pipeline solo, end to end: discover (Google Places) → crawl → **extract with Claude (Batch API)** → **score & rank** → serve a personalized dashboard — **5 states, 47 metros, ~285 cities, 15,000+ vendors**.
- Per-category scoring model blends weighted features with data-sparsity and wedding-relevance gating (learned from real crawl-quality failures); cost-engineered to ~\$0.0025/vendor under Google's free tier. Stack: Next.js 16 / React 19, Neon Postgres, Clerk, Stripe, Vercel.

Mini-Netflix RecSys — Production-Style Recommender *PyTorch, FAISS, LightGBM, FastAPI*

- Two-stage system: PyTorch two-tower retrieval (FAISS ANN, <5 ms) → LightGBM LambdaRank ranking, served via FastAPI at **~45 ms/request** with dynamic batching, backpressure, a Redis feature store (train/serve skew checks), and Prometheus metrics (~3,200 LOC).
- Ran a simulated A/B test with bootstrap CIs and measured a statistically significant NDCG@10 lift from the ranking stage (diagnosed and fixed an unstable relative-lift estimator that produced meaningless CIs).

Token Convergence — Interpretability Research *PyTorch, Transformers*

- Measured per-token, per-layer representation convergence across **Pythia-1.4B/6.9B and Mistral-7B** (32K+ tokens); reported an honest negative result on KV-cache speedups with causal freeze-intervention validation. Drafting a workshop paper.

LLM Plays Pokémon — Autonomous Agent *Claude API, Python*

- Autonomous agent that plays Pokémon FireRed from emulator memory with deterministic BFS navigation and cost-aware Haiku/Sonnet routing — **88% of 69K actions ran as free deterministic steps**, keeping total API spend under \$10 (SQLite-logged token/cost accounting).

Also: kvviz (KV-cache profiler), Parameter Golf (16 MB model-compression challenge), tiny quantized LLMs on Nintendo DS.

PROFESSIONAL EXPERIENCE

Transportation Engineer — Tennessee Department of Transportation

Jan 2020 – Jul 2025, Feb 2026 – Present

- Perform and author traffic safety and operations studies on state highways; review site plans and traffic impact studies for commercial driveway permits and develop conceptual road-safety improvement plans.
- Built a **data-driven crash-risk ranking model** that scored every 4-lane divided-highway intersection lacking a left-turn lane by crash count, AADT, speed limit, and crash rate — surfacing the highest-risk sites for safety investment (queried the TDOT crash database + GIS).
- As Transportation Engineering Specialist (2020–24), helped coordinate **50+ resurfacing and bridge-replacement projects worth \$20M+ annually across 24 East TN counties**; computed quantity estimates and ran inspections for compliance.

Traffic Operations Specialist — Y-12 National Security Complex

Jul 2025 – Feb 2026

- Built a system to log and track workzone traffic verifications and traffic studies for standards compliance and auditability; partnered with developers on a new platform to host, share, and log the department's traffic work.

Traffic Engineer — Felsburg, Holt & Ullevig · Synchro/VISSIM traffic modeling

Jun – Nov 2019